Minseop Choi

AI Systems Engineer - Agentic AI, Model Customization & Production Delivery

Seoul, South Korea | Open to relocation to Tokyo
minsub0922@gmail.com | github.com/minsub0922

SUMMARY

AI systems engineer with **6+ years** across production AI, LLM training, and agentic systems. Translate business requirements into **enterprise agent architectures and production deployments**, from task decomposition and orchestration through model customization, evaluation, observability, and security. Combine hands-on delivery with developer mentorship to turn ambiguous requirements into testable workflows and reusable engineering patterns.

PROFESSIONAL EXPERIENCE

Samsung Research - AX Group *Aug 2026 - Present*
AI/ML Engineer Seoul, Korea
Winmate / Samsung Proposal Studio - Internal B2B proposal platform

- **Designed, launched, and operate** an internal multi-agent proposal platform covering requirement analysis, product/solution recommendations, sales strategy, and editable PPTX generation; won an **internal AX Award**, with estimated workflow effort reduced from **66 to 22 FTE equivalents**.
- Worked directly with business stakeholders to define **PRDs and measurable success criteria**; translated proposal workflows into domain agents and reusable service contracts, using **RAG/knowledge-graph context** and a **CLM-8B decision model** for slide-layout selection.
- Designed **multi-agent orchestration with compressed global and task-local context**, Redis-backed gateway queues for burst traffic, and sandbox isolation/execution policies; validated workflows through **persona-based proposal simulation and Persona QA**.
- Established **Langfuse LLMOps and SigNoz/OpenTelemetry tracing** with custom evaluators, TDD, and contract/regression tests to diagnose service and model failures, validate evidence-grounded outputs, and verify fixes.

Samsung Research - AI System *Sep 2021 - Jul 2026*
AI/ML Engineer Seoul, Korea
On-Device AI - Model training, evaluation, and deployment

- Built **GitHub Actions-based post-training and evaluation automation** across **16 languages and seven tasks**, standardizing task/language-specific runs, benchmark execution, and regression checks.
- Built a **hybrid LLM-as-a-Judge + human-feedback evaluation loop** to identify failure patterns and convert them into data curation, training updates, and repeatable validation.
- Trained and deployed on-device LLMs with **PyTorch and Hugging Face Transformers**: a **4B entity-extraction model using SFT/LoRA**, and a **0.6B multitask model** with **mid-training via knowledge distillation** and **post-training via DPO/SFT**.

SRNote - Self-hosted Agent System (Internal AI Agent Hackathon Award Winner)

- Built a self-hosted knowledge agent integrating **Confluence/GitHub through MCP** for classification, summarization, and **project-scoped hybrid RAG** over notes, enterprise documents, and repository context.
- Applied **NVIDIA NeMoClaw/OpenShell** for per-agent knowledge/tool-access isolation, enforcing personal/enterprise context boundaries while preserving autonomous planning, retrieval, and tool execution.

Hyundai Motor Company *Mar 2020 - Aug 2021*
Cloud Solution Engineer Seoul, Korea

- Handled **OpenStack network incidents** and private-cloud deployment/configuration issues; owned migration from legacy hardware-based routers to software-based routing.
- Used **Ansible** to automate provisioning, health checks, incident-response workflows, and maintenance; standardized operational runbooks for repeatable diagnosis and configuration.

TECHNICAL SKILLS

Agentic AI systems: LangGraph, DeepAgents, FastAPI, Pydantic, Redis; multi-agent orchestration, compressed global/task-local context, hybrid RAG, knowledge graphs, tool calling, domain decomposition, Redis-backed gateway queues, reusable service contracts

Evaluation and testing: Langfuse, GitHub Actions; evaluation harnesses, custom evaluators, automated graders, LLM-as-a-Judge, human feedback, Persona QA, TDD, contract/regression and UI/E2E testing

Production and observability: SigNoz, OpenTelemetry/OTLP, Grafana/Loki; distributed tracing, per-service monitoring, request-ID correlation, failure reproduction, debugging, latency/token analysis, retries, rate limiting, circuit breakers

Enterprise integration and security: MCP, Confluence, GitHub, NeMoClaw/OpenShell; domain-knowledge isolation, agent sandboxing, scoped context/tool access, personal-context sharing controls, RBAC

Model training and customization: Python, PyTorch, Hugging Face Transformers; SFT/LoRA, DPO, mid-training with knowledge distillation, post-training, multilingual evaluation

Infrastructure and delivery: OpenStack, Ansible, Docker, Kubernetes, Linux, PM2/systemd; network incident response, health checks, runbooks, CI/CD, Git

AI coding and developer workflows: Codex, Claude Code, Gemini, GitHub Actions; specification-driven development, coding-agent workflows, task decomposition, reusable engineering patterns, regression-driven iteration

TECHNICAL MENTORSHIP

AI & SW Maestro *2026 - Present*
Technical Mentor South Korea
Appointed by the Minister of Science and ICT, Republic of Korea.

- Guide developer teams through **agent architecture, RAG, model deployment, and evaluation**; translate ambiguous requirements into implementation plans, testable acceptance criteria, and validation workflows.
- Deliver **hands-on technical workshops** on AI coding agents and specification-driven development, explaining architecture choices and helping developers apply repeatable engineering practices.

SCPC AI Challenge *Apr 2026 - Aug 2026*
Problem Reviewer / Evaluator South Korea

- Reviewed AI challenge tasks, test cases, and scoring criteria for correctness, difficulty, and evaluation robustness.

EDUCATION AND LANGUAGES

Korea Aerospace University - Goyang, Korea
Bachelor's degree in Computer Science
Languages: Korean (native); English (TOEIC 915, previously achieved); Japanese (conversational).

SELECTED PUBLICATIONS

- **NanoQuant: Efficient Sub-1-Bit Quantization of Large Language Models.**
  H. Chong, D. Kim, C. Kim, **M. Choi**. *ICML 2026*. arXiv:2602.06694.
- **RaBiT: Residual-Aware Binarization Training for Accurate and Efficient LLMs.**
  Y. You, B. Lee, **M. Choi**, et al. *ICML 2026*. arXiv:2602.05367.